

PushAround: Collaborative Path Clearing via Physics-Informed Hybrid Search

Abstract—The passage of large vehicles in cluttered environments is often blocked by movable obstacles, motivating the use of mobile robot teams to proactively clear traversable corridors. Existing approaches to Navigation Among Movable Obstacles (NAMO) mainly plan abstract sequences of obstacle displacements but often neglect physical feasibility, overlooking robot dimensions, obstacle mass, contact geometry, and required forces, which can lead to strategies that fail in execution. This work introduces *PushAround*, a physics-informed hybrid search framework that incrementally constructs executable clearing plans by coupling geometric connectivity reasoning with short-horizon simulation. Repeated W-clearance connectivity graph (WCCG) queries expose frontier gaps where the vehicle corridor must be widened and rank them by estimated downstream costs, while the search generates direct or recursive pushing candidates – choosing which obstacles to displace, in which directions, and with which contact points and forces – admitting only physics-validated candidates. Additional validation characterizes connectivity-query consistency and sensitivity to physics-model mismatch. The framework is validated by extensive simulations and hardware experiments.

I. INTRODUCTION

In many real-world settings, the passage of large vehicles or transport units is obstructed by movable obstacles such as pallets, boxes, or equipment, motivating the deployment of mobile robot teams to actively clear traversable corridors and enable safe passage. This capability is critical in cluttered and unstructured environments such as warehouses, disaster sites, and dense urban spaces. Conventional path planning methods assume static obstacles and thus fail once direct routes are blocked [1]. Navigation Among Movable Obstacles (NAMO) has been studied as a means of creating new routes by pushing, pulling, or rotating obstacles [2], yet most methods abstract away physical feasibility. Factors such as robot dimensions, obstacle size and mass, contact geometry, applied forces and pushing dynamics are typically ignored, producing plans that are *geometrically valid but physically unrealizable in practice*. As shown in Fig. 1, this difficulty is amplified when a large robot cannot reach obstacle boundaries, while a small robot lacks sufficient pushing force. Cooperative pushing by several small robots is a viable alternative, but introduces strong physical coupling since simultaneous pushes can trigger chained motions of multiple obstacles, making prediction and planning substantially more complex.

A. Related Work

NAMO extends motion planning to cluttered environments by planning obstacle displacements through pushing, pulling, or rotating [2], [3]. Later work considers heuristic search, learning-based local planning, physics-based rollouts, and collaborative pushing [3]–[7]. However, many NAMO formulations simplify robot geometry, actuation limits, object mass, and contact dynamics, which can yield unrealizable plans.



Fig. 1. Collaborative path clearing in both simulation (Top) and hardware (Bottom). Two robots actively push movable obstacles to create a W-clear path for a large vehicle to traverse from start to goal, via a hybrid search over the sequence of obstacles to push and the associated pushing modes.

Collaborative pushing has been widely studied in multi-robot manipulation: foundational work on the mechanics of pushing and limit surfaces [8], [9], cooperative transport with robot swarms [10], multi-robot box pushing with reinforcement learning [11], collaborative pushing of polytopic objects in cluttered scenes [7], and learning-based locomotion [12]. These studies focus on a prescribed object rather than multi-object corridor clearing.

Physics-informed planning incorporates dynamics models, contact reasoning, or simulation feedback into motion generation: learned dynamics for humanoid contact planning [13], multi-contact force control [14], demonstration-guided optimal control over contact modes [15], physics simulation interleaved with multi-agent pathfinding [16], [17], and learned physical priors [18], [19]. These methods nonetheless assume a prescribed manipulation target or few objects, and physical realizability is usually argued rather than validated before execution. Corridor clearing differs on both counts: many movable obstacles are rearranged jointly, and the cleared configuration must remain traversable for a larger vehicle.

B. Our Method

As shown in Fig. 2, this work introduces **PushAround**, a physics-informed framework for collaborative multi-robot pushing that actively constructs traversable corridors for large vehicles in cluttered environments. Its central novelty is a hybrid search that jointly determines which obstacles to displace and how to push them, coupling the high-level sequence of obstacle clearing with the low-level pushing modes that specify contact points, directions, and forces. The framework integrates three components: a W-clearance connectivity graph (WCCG) that certifies vehicle passage under width constraints, a gap-ranking strategy that prioritizes frontier gaps by estimated effort, and pushing-mode generation under geometric constraints. These are encapsulated in a search that incrementally expands candidate clearing plans while validating

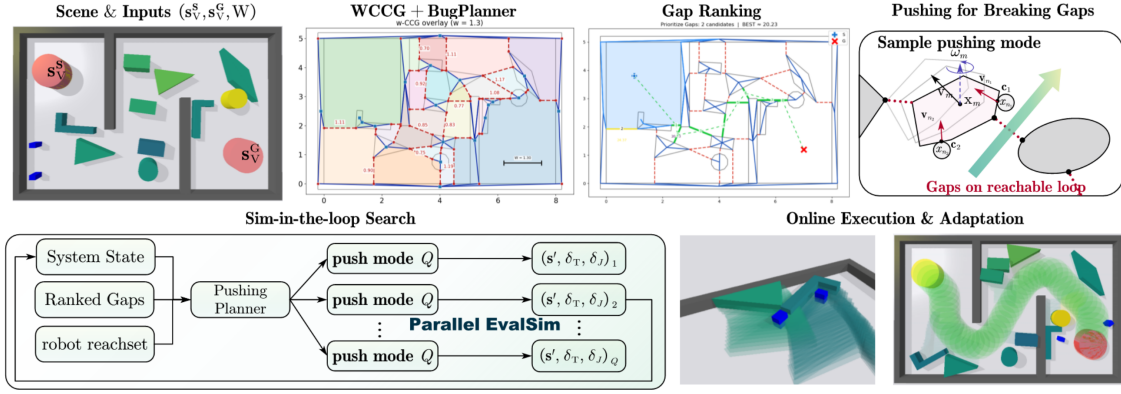


Fig. 2. Overview of the proposed framework. **Top:** Scene inputs, WCCG construction, gap ranking, and sampling of pushing modes for breaking frontier gaps; **Bottom-left:** Sim-in-the-loop hybrid search with parallel validation in simulation; **Bottom-right:** Online execution and adaptation during path clearing.

their physical realizability, so the resulting plans are not only geometrically valid but also feasible under robot dimensions, object mass, contact geometry, and coupled pushing dynamics. Since physics validation depends on model fidelity, robustness under execution-model mismatch is also evaluated.

The contributions are twofold: (I) it introduces a unified framework that couples collaborative pushing with physics-informed verification, producing executable plans for path clearing in cluttered environments; and (II) it shows significant improvements in feasibility, efficiency and scalability over existing NAMO and collaborative clearing methods.

II. PROBLEM DESCRIPTION

A. Workspace and Robots

The workspace is a bounded planar region $\mathcal{W} \subset \mathbb{R}^2$ containing immovable structures $\mathcal{O} \subset \mathcal{W}$ and $M \geq 0$ movable rigid polygons $\Omega \triangleq \{\Omega_1, \dots, \Omega_M\} \subset \mathcal{W}$. Each movable obstacle Ω_m has mass M_m , inertia I_m , frictional parameters and state $\mathbf{s}_m(t) \triangleq (\mathbf{x}_m(t), \psi_m(t))$, with $\mathbf{x}_m(t) \in \mathbb{R}^2$ the centroid position and $\psi_m(t) \in \mathbb{R}$ the orientation; $\Omega_m(t)$ denotes the region it occupies at time t . A team of N robots, $\mathcal{R} \triangleq \{R_1, \dots, R_N\}$, operates as a cooperative unit. Each robot R_i is a rigid body with state $\mathbf{s}_{R_i}(t) \triangleq (\mathbf{x}_{R_i}(t), \psi_{R_i}(t))$ and footprint $R_i(t)$. The instantaneous free space is given by:

$$\widehat{\mathcal{W}}(t) \triangleq \mathcal{W} \setminus \left(\mathcal{O} \cup \{\Omega_m(t)\}_{m=1}^M \cup \{R_i(t)\}_{i=1}^N \right), \quad (1)$$

where $\widehat{\mathcal{W}}(t)$ excludes regions occupied by immovable obstacles, movable obstacles, and robots.

B. External Vehicle and Clearance Goal

An external vehicle $\mathcal{V}$ whose footprint is enclosed by a disk of radius $W/2 > 0$ must navigate within the workspace from a start position $\mathbf{x}_V^s \in \mathcal{W}$ to a goal position $\mathbf{x}_V^g \in \mathcal{W}$. Since movable obstacles may obstruct the way, the clearance goal is to rearrange them so that the workspace contains a passage from $\mathbf{x}_V^s$ to $\mathbf{x}_V^g$ that can accommodate this enclosing disk, i.e., a passage of width at least W . At time t , this is expressed by the admissible reference-center space $\mathcal{F}_W(t) \triangleq \{p \in \widehat{\mathcal{W}}(t) \mid \mathbb{B}_{W/2}(p) \subset \widehat{\mathcal{W}}(t)\}$, where $\mathbb{B}_{W/2}(p)$ is the disk of radius $W/2$ centered at p . The W -clearance condition is then given by:

$$\exists \mathcal{P}_V^W \subset \mathcal{F}_W(t) : \mathbf{x}_V^s \rightsquigarrow \mathbf{x}_V^g, \quad (2)$$

which certifies a collision-free centerline path for any vehicle footprint contained in the enclosing disk.

C. Collaborative Pushing Modes

To enlarge blocked gaps, the robots may actively reconfigure Ω by pushing movable obstacles. For obstacle $\Omega_m \in \Omega$, a pushing mode is denoted by $\xi_m \triangleq (\mathbf{c}_m, \mathbf{u}_m)$, where $\mathbf{c}_m = (c_1, \dots, c_N)$ is an ordered tuple of contact points on $\partial\Omega_m$, and the ordering specifies the robot-to-contact assignment. The vector $\mathbf{u}_m \in \mathbb{R}^{2N}$ stacks the corresponding planar contact forces, expressed in the body frame of Ω_m . For a contact tuple $\mathbf{c}_m$, the set of admissible forces $U_m(\mathbf{c}_m) \subset \mathbb{R}^{2N}$ collects contact forces satisfying the robot pushing limits, robot-object friction, and unilateral contact constraints. The body wrench induced by a mode is defined as:

$$\mathbf{q}_m(\xi_m) \triangleq G_m(\mathbf{c}_m)\mathbf{u}_m, \quad (3)$$

where $G_m(\mathbf{c}_m) \in \mathbb{R}^{3 \times 2N}$ is the contact-to-wrench mapping; and $\mathbf{q}_m(\xi_m) \in \mathbb{R}^3$. The admissible mode set is state dependent as denoted by $\Xi_m(\mathbf{s})$. It contains modes whose contacts are reachable from the current robot configurations, pre-contact configurations are feasible in the current clutter, and forces satisfy $\mathbf{u}_m \in U_m(\mathbf{c}_m)$. Hence, applying a feasible mode $\xi_m \in \Xi_m(\mathbf{s}(t))$ yields the physical transition:

$$\mathbf{s}(t^+) \triangleq \Phi(\mathbf{s}(t), m, \xi_m), \quad (4)$$

where $\mathbf{s}(t)$ stacks all robot and obstacle states. The force vector $\mathbf{u}_m$ serves as a feasibility variable to certify wrench generation under the modeled contact and actuation constraints rather than a directly commanded control input.

D. Problem Statement

The goal is to compute a *hybrid schedule* that reconfigures movable obstacles to admit a W -feasible path for the external vehicle. The schedule is defined as:

$$\pi \triangleq \{(\Omega_{m_k}, \xi_k, \Delta t_k)\}_{k=1}^K, \quad (5)$$

where $\Omega_{m_k} \in \Omega$ is the manipulated obstacle, $\xi_k \in \Xi_{m_k}$ is its pushing mode, and $\Delta t_k > 0$ is the execution duration. The search is guided by

$$\mathcal{J}_{\text{clr}}(\pi) \triangleq T + \alpha \sum_{k=1}^K J(m_k, \xi_k, \mathbf{s}(t_k)), \quad (6)$$

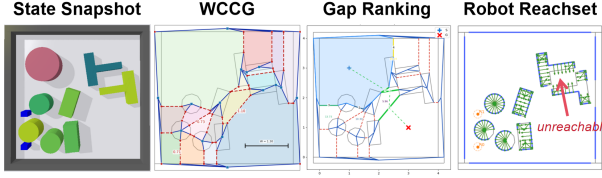


Fig. 3. Core components of the physics-informed hybrid search: WCCG construction, gap ranking, mode generation, and physics validation.

where $T \triangleq \sum_{k=1}^K \Delta t_k$ denotes the robotic clearance duration excluding subsequent vehicle traversal, and $J(\cdot)$ is the simulation-estimated manipulation effort. Vehicle traversal is considered only through the terminal W -clearance condition (2). The constraints (1)–(4) enforce valid pushing transitions. Equation (6) ranks feasible schedules; PIHS uses finite sampling and does not guarantee global optimality of the continuous hybrid objective.

Remark 1. The objective and constraints above couple obstacle selection, pushing-mode generation, and switching in one hybrid planning problem, yielding a combinatorial-continuous search space far richer than classical NAMO formulations with simple objects or hand-crafted contact modes [8], [10], [11]. ■

III. PROPOSED SOLUTION

As shown in Fig. 2, PushAround first uses the W -Clearance Connectivity Graph in Sec. III-A and gap ranking in Sec. III-B to identify and prioritize blocking gaps. It then generates and evaluates physically feasible collaborative pushing modes in Sec. III-C. These components are integrated into the simulation-in-the-loop hybrid search in Sec. III-D, followed by the online execution procedure summarized in Sec. III-E.

A. W -Clearance Connectivity Graph

The feasibility of routing the vehicle reduces to whether a W -width corridor connects $\mathbf{x}_v^S$ and $\mathbf{x}_v^G$. A W -clearance connectivity graph (WCCG) is introduced here to encode obstacle adjacencies under clearance W and enable fast connectivity queries. Conventional clearance checks either rasterize the workspace into an inflated occupancy grid or query sampling-based roadmaps, and must therefore fix a resolution and re-query the map whenever an obstacle moves [20]. Built directly from obstacle geometry, the WCCG introduces no grid-resolution parameter in the repeated connectivity queries used during planning.

1) *Graph Construction:* The WCCG is built by decomposing all clearance-relevant obstacle and workspace boundaries into convex components or segments. From each component C , a centroid node v_c is created. When two components C_u and C_v have closest points $p_u \in C_u$ and $p_v \in C_v$ with distance less than W , bridge nodes are added at p_u and p_v . These bridge nodes are connected by a bridge-bridge edge, annotated with the gap width $w_{uv} \triangleq \|p_u - p_v\|$, and further linked back to their centroids with centroid-bridge edges. Narrow passages with $w_{uv} < W$ are marked as potential bottlenecks. The resulting graph is given by:

$$\mathcal{G}_W \triangleq (\mathcal{V}, \mathcal{E}_c \cup \mathcal{E}_b), \quad (7)$$

where $\mathcal{V}$ are centroid and bridge nodes; $\mathcal{E}_c$ contains centroid-bridge edges; and $\mathcal{E}_b$ for bridge-bridge edges with widths w_{uv} .

2) *Connectivity Criterion:* Once $\mathcal{G}_W$ has been constructed, connectivity queries can be performed without explicitly computing a geometric path. Let $\text{Face}(\mathcal{G}_W)$ denote the set of connected planar faces induced by the embedded WCCG, i.e., the maximal regions separated by obstacle components and WCCG bottleneck edges. The criterion for (2) is given by:

$$(\mathbf{x}_v^S, \mathbf{x}_v^G \in \mathcal{F}_W(t)) \wedge (\exists F \in \text{Face}(\mathcal{G}_W) : \mathbf{x}_v^S, \mathbf{x}_v^G \in F) \quad (8)$$

The criterion follows the standard configuration-space inflation interpretation: expanding obstacles by a disk of radius $W/2$ turns gaps narrower than W into overlapping barriers. Non-convex obstacles are handled through clearance-relevant decomposition, after which WCCG records the separating bottlenecks. The consistency between WCCG connectivity and continuous and rasterized configuration-space references is further evaluated numerically in Sec. IV-A3. A frontier-tracing procedure, similar to the BugPlanner [21], is then used to extract a skeleton Σ as an ordered sequence of centroid and bridge nodes. Starting from $\mathbf{x}_v^S$, it casts a ray toward $\mathbf{x}_v^G$ and follows the encountered frontier loop until either a valid exit is found or a blocking cycle is returned.

B. Ranking of Frontier Gaps

When the condition in (8) fails, the vehicle cannot reach $\mathbf{x}_v^G$ from $\mathbf{x}_v^S$ through a W -clear path $\mathcal{P}_W^G$. In this case, the planner must prioritize *blocking gaps* on the reachable frontier of $\mathcal{G}_W$ according to their predicted downstream cost, and feed the ranked list to the hybrid search as local clearance targets.

1) *Frontier Extraction and Candidate Gaps:* A BugPlanner query on $\mathcal{G}_W$ above either confirms connectivity or returns a counter-clockwise frontier loop $\mathcal{L}$ that separates the two endpoints. The bridge-bridge edges on $\mathcal{L}$ define the first-hop candidate set $\Gamma_{\mathcal{L}} \triangleq \{g_1, \dots, g_K\}$, where each gap g_k is represented by its two closest boundary points, its width $w_{g_k} < W$, and the adjacent obstacles that can potentially be moved. These candidates are then ranked by predicted downstream cost and passed to the hybrid search as local clearance targets.

2) *Evaluation for Immediate Cost:* Each candidate $g \in \Gamma_{\mathcal{L}}$ is evaluated by combining the cost for the robots to reach the gap and the effort required to widen it. Let $\mathbf{s}_{\mathcal{R}}$ be the current robot positions, and $\mathbf{o}_g$ as the outside insertion point of gap g . The resulting one-hop cost is given by:

$$\mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) = \lambda_t \mathcal{C}_t(\mathbf{s}_{\mathcal{R}}, \mathbf{o}_g) + \lambda_p (\mathcal{C}_{dp}(g) + \mathcal{C}_{rp}(g)) \quad (9)$$

where $\lambda_t, \lambda_p > 0$ weight transition and pushing costs, respectively; $\mathcal{C}_t$, $\mathcal{C}_{dp}$, and $\mathcal{C}_{rp}$ denote transition, direct-pushing, and recursive-pushing costs.

3) *Long-term Cost w.r.t. Goal:* Furthermore, to prioritize gaps closer to the goal, a heuristic is added to the evaluation, i.e., $h(g) = \eta \|\mathbf{o}(g) - \mathbf{x}_v^G\|$, with the scaling constant $\eta > 0$. Lastly, a short A*-style presearch is adopted to cross each candidate gap, recompute the next frontier, and continue for a limited width and depth. The *cost-to-connect* is predicted by:

$$\widehat{\text{Cost}}(g) \triangleq \mathcal{C}(g \mid \mathcal{L}, \mathbf{s}_{\mathcal{R}}) + \sum_{g' \in \Pi^*(g)} \{\mathcal{C}(g' \mid \cdot) + h(g')\}, \quad (10)$$

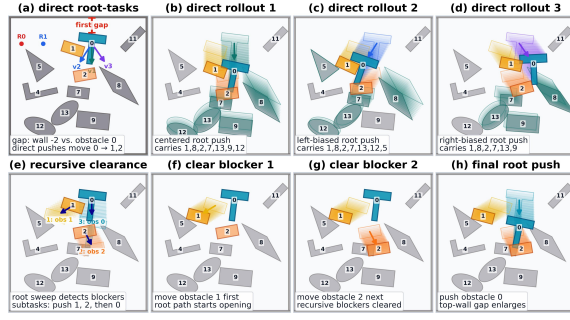


Fig. 4. **Direct** pushing moves a gap-side blocker along a gap-opening direction, whereas **recursive** pushing first clears movable blockers in the swept region and then returns to the root gap-opening task.

where $\Pi^*(g)$ is the sequence of subsequent gaps discovered after virtually crossing g . Consequently, the final output of this module is the ranked list of candidate gaps, i.e., $\text{Rank}(\Gamma_{\mathcal{L}}) \triangleq \text{argsort}_{g \in \Gamma_{\mathcal{L}}} \widehat{\text{Cost}}(g)$, in ascending predicted cost. This ordered set is passed to the hybrid search module, such that only the promising gaps are expanded first.

C. Pushing Mode Generation and Evaluation

For a selected frontier gap g , this subsection constructs a set of candidate pushing strategies $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$, where $\mathbf{v} \triangleq (v_x, v_y, \omega) \in \mathbb{R}^3$ is a short-horizon pushing direction for the movable obstacle Ω_m induced by g , and $\xi \triangleq (\mathbf{c}_m, \mathbf{u})$ is the associated pushing mode in (4). The wrench feasibility encoded in ξ acts as a screening condition rather than a force setpoint: it certifies that the chosen contact configuration can realize the required object motion within the robots' actuation limits, after which the candidate is passed to the physics rollout for validation.

1) *Direct and Recursive Pushing Tasks:* For *direct push*, let Ω_m denote one of the two movable obstacles adjacent to g . For each such obstacle, a small set of candidate body velocities $\{\mathbf{v}_m^k\}_{k=1}^{K_m}$ is sampled with a bias toward enlarging the gap g , while a few angular components are included to account for local geometry. If the swept region of a direct push intersects movable blockers, *recursive push* tasks are required instead: for each out-layer blocker Ω_m , the nominal direction moves it away from both the swept region and the reference gap sequence $\Pi^*(g)$. Recursion therefore creates a bounded dependency chain which prioritizes the final blocker Ω_m , as shown in Fig. 4.

2) *Pushing Mode Generation:* At a search state $\mathbf{s}$, a pushing mode is generated for each candidate direction $\mathbf{v}_m^k$ and obstacle Ω_m . In this subsection, contact tuples, object velocities, forces, and wrenches are expressed in the body frame of Ω_m . For a contact tuple $\mathbf{c}$, its multi-directional force feasibility is:

$$J_{\text{MF}}^m(\mathbf{c}, \mathbf{v}_m^k) \triangleq \sum_{\mathbf{v}_d \in \mathcal{D}_m^k} w_d \min_{\mathbf{u} \in U_m(\mathbf{c})} \|\mathbf{q}_m(\mathbf{c}, \mathbf{u}) + \mathbf{q}_m^\mu(\mathbf{v}_d)\|, \quad (11)$$

where $\|\cdot\|$ is a weighted ℓ_1 wrench norm, the finite set $\mathcal{D}_m^k$ perturbs $\mathbf{v}_m^k$, and $\mathbf{q}_m^\mu(\mathbf{v}_d)$ is the quasi-static friction wrench; the inner minimization is a small LP over the admissible forces $U_m(\mathbf{c})$. A state- and direction-dependent contact set $\mathcal{A}_m^k(\mathbf{s}) \subset \partial\Omega_m$ is drawn from the robot-reachable boundary

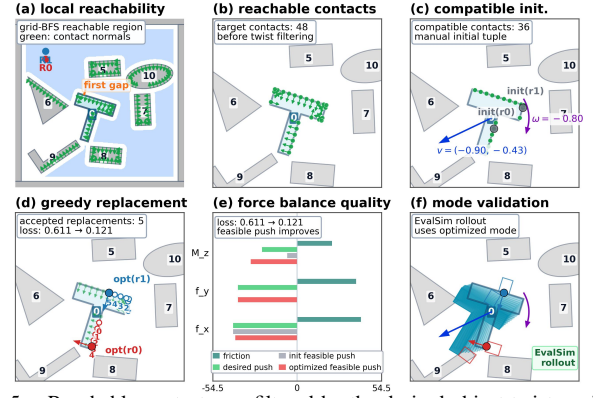


Fig. 5. Reachable contacts are filtered by the desired object twist, assigned to robots, refined by local replacement, and finally validated by simulation.

of Ω_m by velocity–force compatibility filtering (Fig. 5), keeping a contact only if its robots reach collision-free pre-contact poses. Starting from cached layouts or random initializations, the tuple is refined by sequential greedy replacement: at the ℓ -th step, $\mathbf{c}_{r,c}^\ell$ replaces the r -th contact with c , and the best replacement is:

$$\mathbf{c}^* = \underset{\mathbf{c} \in \mathcal{A}_m^k(\mathbf{s})}{\text{argmin}} J_{\text{MF}}^m(\mathbf{c}_{r,c}^\ell, \mathbf{v}_m^k), \quad (12)$$

swept over contact slots until J_{MF}^m no longer decreases. A bounded backtracking variant removes the robot-order sensitivity of this replacement. If no full-team assignment exists for a task, e.g., on small faces or under the pairwise pusher-clearance constraint, one robot is released for that single push (charged a penalty when ranking modes, so full-team modes are preferred) and the full team is available again afterwards. The best tuple over initializations gives $\mathbf{c}_m^{k,*}$, from which $\mathbf{u}_m^k$ follows from the inner minimization in (11), yielding $\xi_m^k = (\mathbf{c}_m^{k,*}, \mathbf{u}_m^k)$; each mode is cached as a warm start. The best few admissible pairs are kept for gap g and passed to simulation; $\mathbf{u}_m^k$ certifies wrench feasibility while hardware tracks the contact geometry from pose/velocity commands.

3) *Pushing Strategy Evaluation:* For a current state $\mathbf{s}$ and a candidate strategy $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$, a short-horizon rollout evaluates the transition in (4) under the selected mode. As shown in Fig. 5, a strategy is accepted only if the robots reach the pre-contact poses, maintain stable contact, and induce obstacle motion above threshold. The simulation returns:

$$(\mathbf{s}', \delta_T, \delta_J) \triangleq \text{EvalSim}(\mathbf{s}, (\Omega_m, \mathbf{v}, \xi)), \quad (13)$$

where $\mathbf{s}'$ is the successor state capturing the resulting robot and object configurations; $\delta_T > 0$ is the time cost; and δ_J is the control effort. The rollout jointly simulates all robots and movable bodies, thereby capturing contact-induced object interactions. For efficiency, the pushing rollout may be bypassed when the mode has been validated before and the swept region of Ω_m is collision-free.

D. Physics-Informed Hybrid Search

The proposed physics-informed hybrid search (PIHS) selects blocking gaps and validates corresponding multi-robot pushing strategies through short-horizon simulation, as shown in Fig. 6 and summarized in Alg. 1. Unlike purely geometric

Evaluation of Pushing Modes via Parallel Simulation:

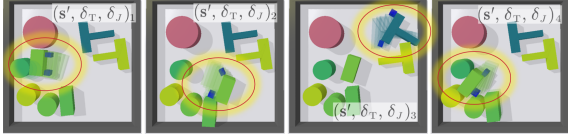


Fig. 6. Sampled push tasks are evaluated via parallel simulation, producing updated states for search expansion.

NAMO planners, PIHS expands only physics-validated strategies through parallel candidate evaluation. It is a feasibility-oriented heuristic search, where Eq. (6) defines the ranking objective for candidate schedules. The search operates on a finite sampled candidate set and therefore does not guarantee global optimality or completeness of the underlying continuous hybrid planning problem.

1) *Tree and Initialization*: The search tree $\mathcal{T}$ holds nodes $\nu \triangleq (s, \pi)$, where s is the system state of all robots and movables in (4) and π the partial pushing strategy in (6). Two global functions are maintained: $\text{Rank}(\nu)$ stores the ranked candidate gaps $\Gamma_{\mathcal{L}_\nu}$ of the frontier loop $\mathcal{L}_\nu$ together with their exploration costs by (10), and $\chi(\nu)$ returns the cumulative execution cost from the root to ν . The root is $\nu_0 \triangleq (s_0, \emptyset)$ with initial state s_0 .

2) *Node Selection*: At each iteration, the node with minimum best-first priority is selected from the queue. Priority balances realized cost and estimated remaining effort:

$$f(\nu) \triangleq \chi(\nu) + \min_{g \in \text{Rank}(\nu)} \widehat{\text{Cost}}(g), \quad (14)$$

where $\chi(\nu)$ is the cumulative execution cost so far, and the second term is the minimum predicted remaining cost among the unexplored gaps of ν . The remaining-cost term is a heuristic and is not required to be admissible. The score prioritizes nodes by realized cost and promising future actions, but it does not provide an A* optimality guarantee.

3) *Node Expansion with Parallel Simulation*: Let the selected node be $\nu \triangleq (s, \pi)$. For each ranked gap $g \in \text{Rank}(\nu)$, its predicted gap sequence $\Pi^*(g)$ in (10) is first derived. Then, the mode optimization by (11) provides a candidate strategy set $\Xi_g \triangleq \{(\Omega_m, \mathbf{v}, \xi)\}$, including both the direct push of the adjacent movable obstacles, and recursive push of out-layer obstacles. Thus, expansion is prioritized first by the gap ordering in $\text{Rank}(\nu)$, and then by the candidate ordering within Ξ_g . Each candidate strategy $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$ is first checked by a geometric quick-pass and, if it passes, evaluated by the short-horizon rollout in (13). A successful evaluation yields (s', δ_T, δ_J) , from which a child node $\nu' \triangleq (s', \pi')$ is created by appending $\tau \triangleq (\Omega_m, \mathbf{v}, \xi)$ to the current partial schedule π . The cumulative cost is updated by: $\chi(\nu') \triangleq \chi(\nu) + \widehat{\text{C}}(\nu, \tau)$, where $\widehat{\text{C}}(\nu, \tau) \triangleq \delta_T + \alpha \delta_J$ is the realized incremental cost. Note that all candidate strategies in the current batch are evaluated *concurrently in parallel*, and a single expansion can produce multiple validated successors.

4) *Termination*: Termination occurs when a node ν reaches a state s for which a W -clear path $\mathcal{P}_V^W$ exists from $\mathbf{x}_V^S$ to $\mathbf{x}_V^G$ by (8). Thus, the schedule π stored in ν is a feasible solution satisfying the terminal clearance condition and the validated pushing transitions. It encodes both the sequence of gaps to

Algorithm 1: Physics-Informed Hybrid Search

Input: $s_0, \alpha, \text{EvalSim}(\cdot), W$ -criterion by (8)
Output: π^*

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1  $\nu_0 \leftarrow (s_0, \emptyset), \chi(\nu_0) = 0;$ 
2 Init  $\mathcal{Q}$  by (14);
3 while not terminated do
4    $\nu \leftarrow \mathcal{Q}.\text{pop\_min}()$  by (14);
5   foreach  $g \in \text{Rank}(\nu)$  do
6     Compute  $\Pi^*(g)$ , generate  $\Xi_g$ ;
7     foreach  $(\Omega_m, \mathbf{v}, \xi) \in \Xi_g$  do
8        $\tau = (\Omega_m, \mathbf{v}, \xi)$  // Direct or recur.
9        $(s', \delta_T, \delta_J) \leftarrow \text{EvalSim}(\nu, \tau)$  by (13);
10      if success then
11         $\nu' \leftarrow (s', \pi \cup \{\tau\});$ 
12         $\chi(\nu') \leftarrow \chi(\nu) + \delta_T + \alpha \delta_J;$ 
13        Add  $\nu'$  to  $\mathcal{Q}$ ;
14      if  $s'$  admits  $\mathcal{P}_V^W$  by (8) then
15         $\pi^* \leftarrow \pi',$  break;
16 return  $\pi^*;$ 

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clear and the pushing strategies that realize them, without implying that $\mathcal{J}_{\text{clr}}(\pi)$ is globally minimal. The objective covers the clearing phase only, since the external vehicle traverses the corridor after clearing completes; traversal time is therefore not part of T .

Remark 2. Efficiency comes from three lightweight heuristics: gap-lookahead reduces unproductive branches, MODE-CACHE warm-starts mode optimization, and the quick-pass rollout bypass skips simple motions whose mode was previously validated, avoiding redundant simulations. ■

E. Overall Analyses

1) *Execution and Online Adaptation*: The hybrid search returns a pushing schedule $\pi = \{\tau_1, \dots, \tau_K\}$, with $\tau_k \triangleq (\Omega_k, \mathbf{v}_k, \xi_k)$. The schedule is executed in a receding-horizon manner: for each τ_k the robots reach the mode ξ_k and follow the pushing direction $\mathbf{v}_k$ from the current state, as in (13). On the hardware the optimized contact forces certify feasibility rather than being commanded: a mode is retained only if its object wrench lies within the robots' actuation limits, and over each control interval EvalSim predicts a short forward object-motion reference (the expected displacement and rotation of the object), which the velocity-level controller tracks with motion-capture feedback, and each executed push returns an observed state that re-enters the planner. During transitions, robots are reassigned by preserving the circular order of contact points along the obstacle boundary [7], reducing path crossings under sufficient clearance. After each push the state is updated from observation, the WCCG is rebuilt from the current configuration, and execution continues until a W -clear path $\mathcal{P}_V^W$ connects $\mathbf{x}_V^S$ and $\mathbf{x}_V^G$. Replanning is triggered by observed execution discrepancies – failed contact establishment, stalled pushing, or invalidated WCCG connectivity – in which case Alg. 1 is called from the current state to derive the updated schedule.

2) *Computational Complexity*: Given M movable obstacles, the computational cost of one node expansion in Alg. 1 is dominated by candidate generation and physics validation. The

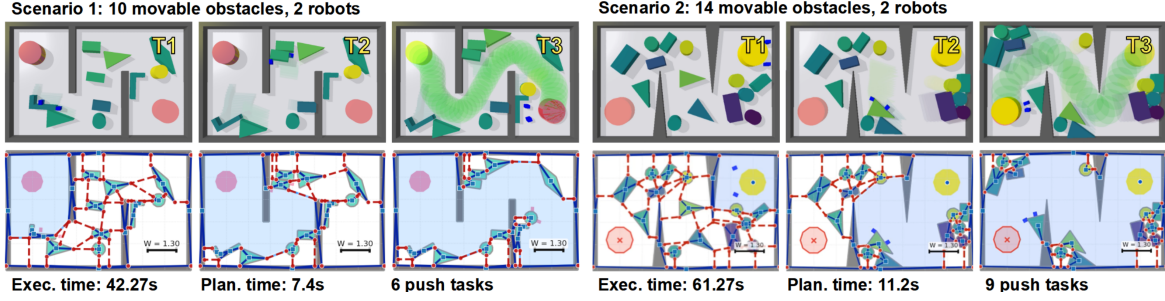


Fig. 7. Execution and planning alignment in Scenario 1 (Left) and Scenario 2 (Right). **Top:** simulated snapshots during path clearing. **Bottom:** corresponding WCCG overlays with the start face in blue. The executed pushes progressively expand the reachable W -clearance region until the goal is connected.

TABLE I
WCCG CONNECTIVITY VALIDATION AND MEAN RUNTIME.

Type	CC=FR	FP/FN [†]	Time (ms)		
			CC	FR	WCCG
Convex	100/100	0/0	1.23	282.9	0.86
Non-convex	98/100	0/0	1.51	204.1	1.30
Mixed	98/100	0/0	1.33	242.4	1.05
Total	296/300	0/0	1.36	243.1	1.07

[†]WCCG errors with respect to the continuous configuration-space reference (CC). All runtimes are measured in one harness over the same 300 scenes, with WCCG served by its compiled backend;

WCCG update and gap ranking incur $\mathcal{O}(M \log M + |\mathcal{V}| + |\mathcal{E}_c| + |\mathcal{E}_b|)$ and $\mathcal{O}(|\Gamma_{\mathcal{L}_v}| \log |\Gamma_{\mathcal{L}_v}|)$ costs, respectively. The dominant simulation cost $\mathcal{O}(\sum_{g \in \text{Rank}(\nu)} |\Xi_g| (C_{\text{LP}} + \rho_{\nu} T_{\text{sim}}/n_w))$ grows with the number of validated candidates and is reduced by the n_w parallel workers, so scalability is set by validated candidates rather than by the enumeration of contact assignments. Robot-team size mainly affects mode generation: the greedy stage evaluates at most $N|\mathcal{A}|$ single-contact substitutions instead of the exhaustive $|\mathcal{A}|^N$ assignment, and when it fails, a bounded-backtracking search over robots and candidates expands at most $B=2000$ partial assignments, a fixed budget independent of the team size; the subset fallback repeats this search for at most one excluded robot. The number of LP feasibility checks per task is therefore at most $(N|\mathcal{A}|+2B)$, each over the $2N$ admissible contact forces, rather than growing as $|\mathcal{A}|^N$.

IV. EXPERIMENTS

Numerical experiments are conducted to validate the proposed PIHS algorithm. The algorithms are implemented in Python3 and PyBullet [22] on a computer with an Intel Core i9-14900KF and an NVIDIA RTX 4080. Videos are provided as multimedia attachments.

A. Numerical Simulations

1) *Setup and Configuration:* The simulation runs at a step size of $\Delta t = 1/240$ s with a control period of $1/40$ s; robots are disk or box pushers consistent with the W -clearance definition, and movable obstacles have masses sampled from $[5, 15]$ kg. A trial succeeds when the external vehicle reaches its goal through a W -clear path. The nominal scenario is an 8×5 m workspace with bottleneck-forming bar obstacles, where two robots clear 10–15 movable obstacles under minimum-separation constraints. Unless stated otherwise, comparisons and ablations use Scenario 1 (10 movable obstacles, 40 trials), and Scenario 2 (14 obstacles) provides an additional cluttered

TABLE II
PERFORMANCE COMPARISON AND ABLATIONS ON SCENARIO 1 OVER 40 RANDOMIZED TRIALS.

Method / Variant	Succ.(%)	PT (s)	ET (s)	#Sims	#Pushes
DFS-WCCG	25.0	>100.0	51.2	>500	8.0
SL-Push (offline)	62.5	0.06	44.1	0.0	7.3
SL-Push (sim)	75.0	18.2	64.6	20.0	10.2
Rec-NAMO	37.5	13.3	42.4	0.0	7.8
PushAround (ours)	97.5	10.3	28.6	121.5	6.0
w/o gap-lookahead	82.5	16.9	35.2	178.3	7.2
w/o mode-cache	92.5	17.1	33.4	164.6	6.5
w/o quick-pass	90.0	23.6	41.9	208.2	6.1
w/o reinsertion	85.0	12.8	31.1	151.7	7.8

case. All methods share the robot footprints, obstacle configurations, collision and reachability checks, and execution simulator, with the same per-trial timeout of 800 s and simulator cap of 20,000 control steps; unsuccessful trials are assigned the 800 s cap in the planning-time statistics.

2) *Results:* Two representative simulation runs are shown in Fig. 7. In Scenario 1 the start and goal lie in different W -clearance faces, so the planner clears the blocked passage with 6 pushes in 7.4 s and execution finishes in 42.3 s. Scenario 2 requires a 9-push plan in 11.2 s and 61.3 s of execution. In both cases the executed pushes progressively enlarge the reachable W -clearance region until a complete W -clear path is found, as shown by the expanding blue start face in the WCCG overlays.

3) *WCCG Connectivity Validation:* WCCG is compared against two references on 300 random scenes covering convex, non-convex and mixed obstacles (Table I). The continuous reference (CC) inflates obstacle and workspace boundaries by $W/2$ and queries connectivity with a fine continuous collision check at 0.01 m oracle resolution instead of a fixed grid [23]; the rasterized reference (FR) discretizes the same workspace at 0.01 m on an occupancy grid. CC and FR disagree in 4 of the 300 cases, indicating sensitivity of the raster reference near narrow or tangential contacts. WCCG matches CC on all 300 scenes with zero false positives and zero false negatives. A query costs 1.07 ms against 1.36 ms (CC) and 243 ms (FR), while additionally supplying the gap graph needed by PIHS.

4) *Comparisons:* Four baselines run on Scenario 1 over 40 randomized trials under the same W -clearance criterion, object distribution and simulator: **DFS-WCCG** (WCCG-guided search with fixed pushes), **SL-Push** (offline and simulation-validated variants), and **Rec-NAMO** (recursive NAMO) [3]–[5], [16]. Table II shows that PushAround reaches the highest



Fig. 8. Failure cases by baselines Rec-NAMO, SL-Push and DFS-WCCG.

TABLE III
SCALABILITY AND ROBUSTNESS UNDER OBSTACLE DENSITY,
ROBOT-TEAM SIZE, AND EXECUTION-SIDE PHYSICS MISMATCH.

Case	Trials	Succ.(%)	PT^* (s)	ET^* (s)
<i>Obstacle density</i> (PushAround, $N=2$, nominal)				
$M = 15$	10	100	2.5 ± 0.9	37.1 ± 21.0
$M = 30$	10	100	11.7 ± 3.7	100.9 ± 34.7
$M = 45$	10	100	53.4 ± 24.5	272.7 ± 97.4
<i>Robot-team size</i> ($M=30$, nominal; 10 trials per cell)				
PushAround, $N=3$	10	100	30.0 ± 12.7	97.3 ± 42.1
PushAround, $N=4$	10	100	29.2 ± 8.9	110.1 ± 33.1
SL-Push, $N=3$	10	80	30.0 ± 46.3	193.3 ± 111.3
SL-Push, $N=4$	10	80	23.4 ± 18.2	171.7 ± 61.2
DFS-WCCG, $N=3$	10	100	142.6 ± 183.1	178.1 ± 72.1
DFS-WCCG, $N=4$	10	90	101.9 ± 103.9	175.9 ± 113.6
<i>Physics mismatch</i> ($M=30$, $N=2$)				
Nominal	10	100	16.8 ± 3.4	104.7 ± 37.4
Mass $\times 0.6-1.4$	40	100	16.8 ± 5.4	105.9 ± 44.6
Lateral fric. $\times 0.6-1.4$	40	97.5	16.5 ± 5.7	114.0 ± 48.5
Spin fric. $\times 0.6-1.4$	40	97.5	15.2 ± 3.6	100.1 ± 35.7
Force limit $\times 0.6-1.4$	40	95.0	18.9 ± 10.9	115.9 ± 58.5
All perturbed	160	97.5	16.9 ± 7.0	108.9 ± 47.4

Each perturbed row scales one parameter; the others stay nominal.

success rate (97.5%) and the lowest combined planning-and-execution time (38.9 s). DFS-WCCG needs more than 100 s of planning (exponential branching); SL-Push (offline) plans in 0.06 s but many displacements fail in execution (62.5%); SL-Push (sim) reaches 75.0% at 18.2 s planning; and Rec-NAMO usually fails to build a connected W -clear corridor in dense clutter, as illustrated in Fig. 8.

5) *Ablation Studies*: Four internal ablations are conducted: **w/o gap-lookahead** (gaps ranked without downstream costs), **w/o mode-cache** (no warm start from validated modes), **w/o quick-pass** (all candidates evaluated by rollout), and **w/o reinsertion** (failed nodes not reconsidered). Table II shows that gap-lookahead and reinsertion mainly improve robustness (success drops by more than 10% when either is removed), whereas mode-cache only lowers success to 92.5%; quick-pass affects efficiency, with simulations rising from 121.5 to 208.2 and planning from 10.3 to 23.6 s.

6) *Scalability and Robustness Analysis*: Generalization is further evaluated in the same 8×8 m workspace by scaling the obstacle density M , the robot-team size N , and the execution-side physics parameters (Table III and Fig. 10). (I) **Obstacle density**: with two robots and 10 trials per density, PushAround maintains a 100% success rate for all $M \in [15, 45]$ at 2.5–53.4 s of planning, whereas at $M=45$ the external methods solve none of the trials and the no-recursion ablation solves 7/10. (II) **Robot-team size**: at $M=30$, PushAround solves all ten trials at $N=2, 3$, and 4 after the shared mode-generation refinements, which reduce robot-order sensitivity and allow a temporary per-task robot release when no full-team assignment

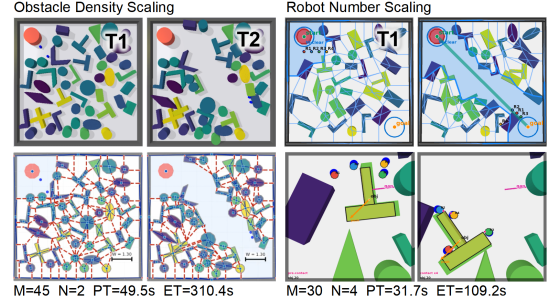


Fig. 9. Scalability of the clearing process. **Left**: obstacle-density scaling ($M=45$ objects, two robots): snapshots during the clearing (top) and WCCG overlays with the start face in blue (bottom). **Right**: robot-team scaling ($M=30$ objects, four robots): initial and final states (top) and zoomed views of one push before contact and during execution (bottom).

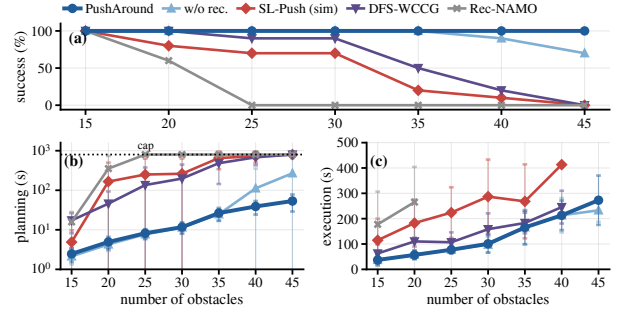


Fig. 10. Obstacle-density scaling (8×8 m, 10 independent seeds per density). (a) Success rate; (b) mean planning time, with unsuccessful trials capped at the 800 s budget (dotted line); (c) mean execution time of the successful trials (Rec-NAMO has none beyond $M=20$). Error bars denote one standard deviation across seeds.

exists, with a ranking penalty preferring full-team modes. At $N=4$, PushAround uses 29.2 s of planning and 110.1 s of execution, compared with 101.9 s and 175.9 s for DFS-WCCG. (III) **Physics mismatch**: the planner keeps the nominal model and only the execution simulator is perturbed, one parameter at a time. PushAround succeeds in 156 of 160 perturbed trials (97.5%). Under the same $\pm 40\%$ perturbations, SL-Push (sim) succeeds in 13/40 trials and DFS-WCCG in 23/40 trials; their mean planning times are 31.5 s and 229.1 s, respectively, compared with 18.4 s for PushAround.

B. Hardware Experiments

1) *System Description*: Experiments use a 5×6 m workspace of interlocking 0.6×0.6 m foam mats (Fig. 11), 2 UGVs and 6 cardboard-box movable obstacles. Robots and movables each carry three to four motion-capture markers whose global poses are streamed to PyBullet in real time. The two Mecanum UGVs are velocity-controlled, and the optimized contact forces are used only for feasibility evaluation rather than direct motor control. Motion capture provides the current robot and object poses, compared with the successor state predicted by EvalSim: a push triggers replanning only when the position error exceeds 0.3 m or the orientation error exceeds $\pi/4$, a loose threshold that tolerates slight over-push or off-axis push. Movables are randomly placed for each trial.

2) *Results*: Two representative real-world runs are shown in Fig. 11. In Scenario 1, PIHS breaks five candidate gaps within 6.5 s using 12 node expansions, 49 visited snapshots and search depth 6; the four-task schedule first rotates the yellow

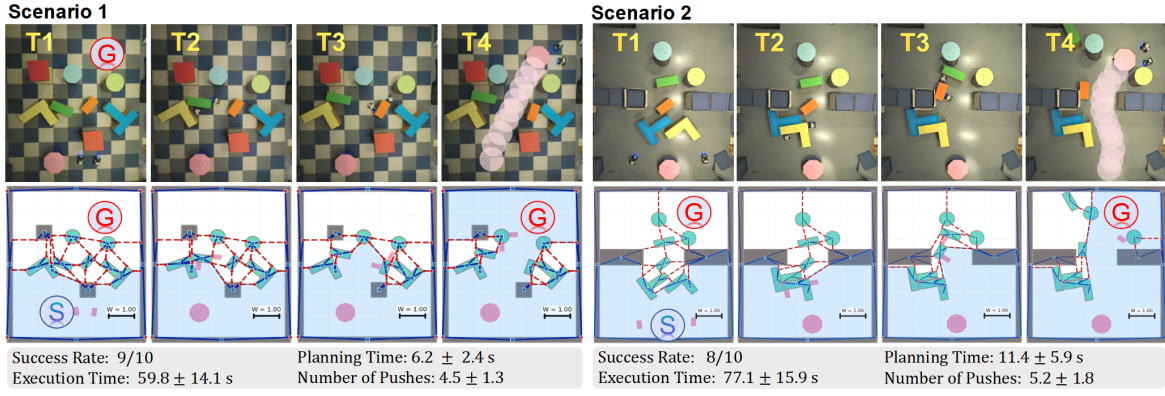


Fig. 11. Real-world pushing experiments with execution-planning alignment (over 20 runs and 2 scenarios). **Top:** snapshots of two robots pushing obstacles in a 5×6 m workspace with movable objects and fixed boundaries. **Bottom:** L-shaped object to open a gap near an immovable obstacle and then separates the green and orange objects until a W -clear path is established at $t = 52$ s. In Scenario 2, PushAround succeeds in 8/10 trials with an average of 5.2 pushes and 11.4 s of planning; it first selects a direct push on the blue L-shaped object, which also moves the adjacent T-shaped block and opens the first gap into the narrow passage. This neighboring-object motion is not a separately encoded action but an emergent outcome of the coupled rigid-body rollout in EvalSim, whose snapshot is used to update the WCCG and continue replanning. Execution takes 77.1 s on average, with 15.9 s of deviations from direction-dependent force limits, drifting and Mecanum-wheel slippage, all compensated by online replanning.

V. CONCLUSION

This work presents PushAround, a multi-robot framework for path clearing in unstructured environments that jointly plans obstacle selection, contact points and pushing forces through a hybrid search, and demonstrates hardware feasibility with state-feedback replanning. Future directions include obstacle-state estimation without external sensing, uncertain masses, and limited robot visibility.

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